

Does the scholarly communication system satisfy the beliefs and aspirations of new researchers? Summarizing the Harbingers research

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Abstract

A study from the Harbingers research project provides a comprehensive assessment of the main features of the scholarly communications system as viewed by early career researchers (ECRs) in the final year of the study (2018). Aspects covered are: discovery and access, authorship practices, peer review, publishing strategies, open access publishing, open data, sharing, collaboration, social media, metrics, impact, reputation, libraries, publishers, and scholarly transformations. Nearly 120 science and social science researchers from seven countries were questioned about these 16 aspects. It was found that some scholarly features work well for ECRs, and in this category can be included: discovery and access, authorship practices, sharing, collaboration, and publishers. Reputation, publishing strategies, and impact are more problematical, and they, in turn, cause tensions regarding some other factors – social media, open access, and open data. Of the rest, libraries are largely invisible, and ECRs have conflicting views concerning ethical behaviour. Few envisage that transformational change will take place in the next 5 years.

Keywords: early career researchers, scholarly communications, publishing behaviour, reputation

INTRODUCTION

If we genuinely want to know what the future of scholarly communications holds, instead of listening to vested interest groups, such as publishers and scholarly platform holders, we need to check with the new generation of researchers – early career researchers (ECRs), who are going to be tomorrow's leading

professors, influencers, and decision makers. This is especially important because they bring with them their millennial beliefs of openness, sharing, and transparency (Burstein, 2013), and they are also on the scholarly frontline, heavily engaged as they are with the scholarly communication tools of the trade.

It has been long speculated that the new wave of researchers will start, or have started, changing the existing scholarly

Key points

- Much of the scholarly communications system works well for early career researchers (ECRs), especially discovery/ access and the sharing and collaborating components.
- Reputation, publishing strategies, and impact are more problematical, with ECRs believing things could be improved.
- Attitudes towards social media, open access, and open data are positive in theory, but practices lag behind because of the dictates of the reputational system, especially paper publishing.
- Libraries are clearly languishing in many countries and are in danger of being disintermediated by the likes of scholarly social networks.

communications system in order to accommodate their millennial beliefs (Anderson & Rainie, 2010; Taylor & Keeter, 2010). We have shown elsewhere in this journal that, while changes over the period 2016–2018 have indeed taken place quickly, they are only taking place with respect to some scholarly features and activities, such as collaboration, social impact, and social media (Nicholas *et al.*, 2019). What we have not done, however, which we intend to do here, is to take stock of each and every one of the major scholarly features (16 in all) that we investigated during the whole project with respect to their perceived value and use by ECRs as of the most recent year of the study (2018). Such an analysis enables us to demonstrate how they measure up, whether they are being reformed, and – if so – how.

LITERATURE REVIEW

It makes a good deal of sense to put millennials at the centre of investigations of scholarly communication behaviour as potential harbingers of change as this generation has been shown to be ‘confident, self-expressive, liberal, upbeat and open to change’ (Burststein, 2013; Pew Research Center, 2010a, 2010b, 2018). However, there have been few studies that investigate junior scholars’ information behaviour patterns with the express aim of establishing whether they are going to be change agents. From the studies undertaken, however, the evidence suggests that ECRs’ attitudes and behaviour tend to comply with extant norms so that any changes discerned are more gradual developments rather than disruptions. This could possibly be because their status as scholarly apprentices mean they are expected to act conventionally (Laudel & Gläser, 2008) and have been found to hold the traditional peer reviewed journal with a high Impact Factor to be the most sought-after outlet for publishing outcomes (Nicholas *et al.*, 2015; Tenopir *et al.*, 2016).

ECRs’ attitudes and practices can be quite conservative as their information-seeking behaviour often shows. Thus, in today’s digital, ever more expansive, and increasingly open and accessible world of

scholarly resources, searching for information has never been easier. Yet, Spezi (2016), found that the whole range of information-seeking behaviours of novice researchers – initiating a search, constructing search strategies, and locating and evaluating the identified sources – follow a steady trend, with only subtle changes from the past, particularly in the use of social media and networking sites. Indeed, the shift towards their greater use of novel ways and means of locating information remain subtle to this very day as the findings of the latest cycle in the series of Ithaka S+R US Faculty Surveys show: while it is the younger researchers who assign the greatest importance to utilizing academic social networks for keeping up with relevant scholarship, the differences among the age cohorts are not very marked (Blankstein & Wolff-Eisenberg, 2019).

As subtle and scattered as the discernible changes in their attitudes and practices may be, there are still indications that, in some areas, today’s ECRs are prepared to stray from the well-trodden scholarly path, especially when these are in line with their millennial beliefs and/or are promising in terms of career advancement and reputational gains. Open access publishing, as part of the wider ideology of Open Science, is certainly a case in point, said to have a host of societal and individual-level benefits (Allen & Mehler, 2019; Enago Academy, 2017; McKiernan *et al.*, 2016). Indeed, junior researchers have been found to manifest much more positive views of open access than their senior counterparts (Nicholas *et al.*, 2015; Ruiz-Pérez & Delgado-López-Cózar, 2017; Tenopir *et al.*, 2016), to the extent that they are even more interested than their older colleagues in replacing the traditional subscription-based system with an open access one (Blankstein & Wolff-Eisenberg, 2019; Wolff, Rod, & Schonfeld, 2016). However, as Segado-Boj, Martín-Quevedo, and Prieto-Gutiérrez (2018) found, it is more nuanced than that. Early career academics among their respondents, who were contributors to Spanish scholarly journals, thought to a greater degree than other age cohorts that open access increases the visibility of articles and citations to them. However, all in all, greater percentages of older respondents and respondents with more years of experience recognized the advantages of open access, but then, above average percentages of the same groups also agreed with its drawbacks.

Beyond that, practice plainly lags behind theory, with ECRs reluctant to carry over their manifestly receptive attitudes towards open access to their actual behaviour as publishing authors. Thus, they have been found to be less likely to view open access characteristics of journals as highly influential in publication decisions (Blankstein & Wolff-Eisenberg, 2019; Tenopir, Dalton, *et al.*, 2016) or, indeed, actually publish in open access journals (Wiley, 2013; Zhu, 2017). True, Rodríguez (2014) did find that the younger age brackets among her respondents had a higher percentage of open access publishing history, but then, they also tended to be tenured so that they could better afford to act upon young people’s aspirations to openness.

METHODOLOGY

Full details of the project’s methodology and the characteristics of the ECR sample can be found in a recently published article in

this journal (Nicholas *et al.*, 2019) and also on the project's Harbinger's website (CIBER Research, 2018; <http://ciber-research.eu/harbingers.html>). In this paper, we provide the broad features of the methodology and also additional methodologies special to the analyses conducted in this paper.

From 2016 to 2018, more than 100 ECRs were subject to annual, in-depth, semi-structured interviews about their scholarly communication beliefs and behaviours. Interviews were mainly conducted, remotely (Skype or telephone) or face-to-face, by domestic university researchers in their home countries and in their own languages. The exception was the USA, which was also covered by the UK interviewer who was conversant with the US scholarly environment. The interview schedule was sent to interviewees ahead of time and contained around 60 mainly open-ended questions. Interviews were not recorded because of cultural sensitivities, and notes were taken instead. All interviewees provided informed consent to participate in the study. Guarantees of privacy, security, and anonymization were given to them all. Transcripts were returned to interviewees in order to validate the data. Transcripts were translated into English for all non-English-speaking countries and manually coded using a heuristic approach and a standardized thematic framework (http://ciber-research.eu/download/20160916-Harbingers-research_instruments.pdf). Selective data were then transferred to spreadsheets for further analysis.

A convenience sample of 116 ECRs was initially derived (Table 1), a number dictated by available funding and the resource-intensive, longitudinal nature of the study. The characteristics of the sample were shaped by: (1) the main funder's (the Publishing Research Consortium's) subject and geographical interests and (2) the availability and co-operation of interviewers on the ground. The study was international with ECRs recruited from seven countries – China, France, Malaysia, Poland, Spain, UK, and USA. Interviewers for the case study countries were provided with a recruitment quota. Within these parameters, the aim was to recruit a sample that would be around two-thirds sciences and one-third social sciences (reflecting the larger numbers of ECRs in science), be reasonably balanced in terms of gender, include researchers from a mixture of universities and some research groups outside universities, and feature ECRs in their 20s and 30s. ECRs were approached via publisher and learned society lists and research networks within individual countries.

By the end of the project, the original panel of 116 fell to 103, largely because of ECRs leaving research as a career. ECRs who moved to a tenured position or moved to another research job elsewhere continued to be interviewed. In addition, a few ECRs stopped co-operating because of job and time issues.

The original interview schedule of more than 60 questions covered 23 aspects of ECRs' work life and their scholarly communications. See Nicholas *et al.* (2019) for a full list of aspects. However, for the purposes of this paper, we will confine ourselves to questions on just the scholarly communication aspects: (1) discovery and access; (2) authorship practices; (3) peer review; (4) publishing strategies; (5) open access publishing; (6) open data; (7) sharing; (8) collaboration; (9) social media; (10) metrics; (11) impact; (12) ethical practices; (13) reputation; (14) libraries; (15) publishers; and (16) scholarly transformations, which we deal with in the conclusions.

National interviewers went through the coded interview data for each of these scholarly aspects to assess how ECRs were dealing with them, what was working well and not so well, and if things are being reformed or reshaped, and these data were coded and transferred to a spreadsheet for further analysis. The national interviewers were all conversant with the educational and societal impact of scholarly communications and separately applied some further context and explanations to the interviewee data. These data were then assembled by the principle investigator, and broad patterns were established in order to give the broad lie of the land.

RESULTS

Discovery and access

Discovery and access are core scholarly activities that work well for ECRs, are seen to be changing in the right direction (that of a borderless, open information environment), and attract very little in the way of negative comment.

Discovery and access are scholarly activities about which ECRs are very familiar, undertake easily and as a matter of fact. While there are myriad discovery tools available to them, ECRs are happy to settle with a small subset. The current set-up, which for the most part consists of Google, Google Scholar, Scopus, Web of Science (not so much the latter two in the cases of USA and UK ECRs), and – for the life and medical sciences – PubMed.

TABLE 1 Characteristics of the original early career researcher panel (N = 116).

Country	No. of ECRs	Social Science (%)	Science (%)	Male (%)	Age 20s (%)	Age 30s (%)	Post-doc (%)
China	13	31	69	54	46	54	92
France	14	21	79	64	64	36	100
Malaysia	12	42	58	50	0	100	100
Poland	10	20	80	60	40	60	50
Spain	18	22	78	56	39	61	72
UK	21	38	62	62	24	76	67
USA	28	21	79	61	29	71	64

Clearly gaining ground, and being added to this list, are ResearchGate and Sci-Hub, with the latter thought by some ECRs to have a simple interface and to provide a good user experience. It is their ubiquity, simplicity, and efficiency that holds the Google products in such high esteem. Google Scholar tops the popularity stakes everywhere, even in China where it is banned. Google itself is also thought to be a very useful tool, even for specialist medical and biomedical scientists in cases where they are not certain what they are looking for. Because of bans on Google products in China, other services, such as CNKI (China National Knowledge Infrastructure, the largest academic online library in China) are popular, although many ECRs manage to circumvent the bans. The interplay between platforms was portrayed by one Chinese ECR who said: 'Yes, first I use Google scholar to do a general search, then in order to be more precise I use WoS and Scopus, usually WoS, but if I want to find a particular author, I use Scopus rather than WoS'.

ECRs are then very satisfied using this battery of research engines and rarely start a search with the library. The starting point for a search in fact, in addition to the classic citations in papers, is often a reference obtained from the social media (e.g. Twitter) and from colleagues. Essentially, they Googled for things they had seen mentioned elsewhere. The improved networking and collaborating opportunities proffered by social media (and much appreciated by ECRs) is clearly making its presence felt on core scholarly activities. It would seem that discovery and access are becoming more personal and social.

ECRs are conscious, of course, of the fact that the full text of an article, generated, for instance, by a search of WoS, is frequently furnished courtesy of library subscriptions. If the library does not provide access to that article, they fall back to Sci-Hub, PubMed Central, ResearchGate, friends, and colleagues and – seemingly as a last resort – interlibrary loan (an administrative process disliked by some ECRs).

There was a surprising lack of complaints about these core activities, but there was one, albeit one not frequently encountered, and that is not being able to obtain the full text of an article. Very pressured lives and short-term attention spans mean waiting is not acceptable, library loans “a bore”, and reference databases are unpopular. 'Provide me with direct access to papers from WoS, Scopus and PubMed' is the general cry. It is possible that the lack of complaints is because Sci-Hub has become a long stop and because ECRs largely came from the top universities and those possessing generous big deals (especially so in the UK and USA). As millennials, ECRs, as we shall learn, are very much in favour of open access, and this would help solve problems of obtaining the full text and those associated with remote logging on, but interestingly, this was not mentioned much in questioning, although it was mentioned a little more so in the case of universities with less-funded libraries.

Authorship

Authorship is both an activity of central interest to ECRs and one about which they exhibit reducing concerns over the duration of the project.

The ECRs' overarching and universal goal is to publish many papers and obtain a prominent position in the author list or be the corresponding author (especially important in China where it obtains a reputational reward). For these reasons, authorship is of much concern and can be an area of much controversy with, for instance, academic staff being given authorship and rank when they are not significantly involved in the paper and those who contribute greatly to the paper not being given their credit in terms of author rank. One ECR summed it up like this: 'young scientists are dependent on the “graciousness” of the older professors and it usually ends with the young scientist doing the whole job, and the “co-author”-professor, signs it and takes the laurels.'

However, over the period of the project, it has become less of an issue as ECRs have developed an understanding of the underlying rules of the game, and complaints are now few and far between. As an illustration, the position of names on papers, frequently decided upon by the research supervisor or head of project, which exercised some ECRs in the early stages of the project (Nicholas *et al.*, 2017), no longer exercises them, and with their growing maturity, it is not unusual for them to be first authors. The understanding developed seems to come from the research group and not the institution.

Peer review

Peer review, perhaps surprisingly, given its notoriety, obtains relatively few criticisms from ECRs.

Virtually all ECRs have had papers reviewed, and many have been reviewers (Rodríguez-Bravo *et al.*, 2017), and the system appears to work well for them or as well as might be expected given the natural tensions associated with evaluation. They much prefer double-blind peer review for the anonymity it affords them, which is especially important because it is believed that they will be penalized because of their juniority. There has, however, been a recent stirring of interest in open peer review. 'The open peer review system can prevent reviewers being too hard (and dismissive) when reviewing a paper and so help improve the article more.'

ECRs prefer publishers to organize peer review (even though, in actuality, it is the Editors who do it), partly because they think the publishers do a good job and partly because they do not like the alternatives – funders or societies who they believe might have vested interests or political agendas. The criticisms they have about the process concerns the fact: (1) that reviewing takes too long; (2) reviewers are not always informed about the topic they are commenting on, not being a specialist or an expert; and (3) they would like more reputational credit for it and, in the case of those aware, appreciate the efforts of Publons (<https://publons.com>) in trying to deliver on this.

The message coming from ECRs is that it works but needs a little tinkering, starting perhaps with the suggestion that 'there needs to be some assessment of the quality of the reviewer'. This is clearly the view of the digital consumer, attuned to rating everything they do.

Publishing strategies

The traditional orthodoxy of publishing in high impact factor journals, reinforced as it is by an unbending reputational and appraisal systems, shows few signs of giving ground to social media and the open science agenda.

Publishing papers in high impact factor journals is very much the reputational concern for ECRs and, of course, their tenured colleagues as well. There is much pressure placed on them to publish in high (and ever higher) impact factor journals, and this is enforced by ministerial and institutional lists of accredited or acceptable journals, which are on the rise as countries (e.g. Poland, Russia, Malaysia) try to raise their rankings in the global top university lists. These prescribed lists, which can run for pages and pages, further cement the dominant scholarly position of the (international) journal and demote the value of monograph and conference paper publishing (generally excluded from these lists), which is believed to disadvantage certain disciplines, such as the arts and humanities. Of course, with these lists freedom of choice goes out of the window, which is something ECRs are uncomfortable with.

For ECRs, however, it is very much a case of accepting this fact rather than liking it. Ideally, however, they would like to do all these things in the sense that they want to publish open access in a high impact factor journal and one that reached their preferred audience. But if this cannot be reconciled – and it is difficult for most of them – the impact factor inevitably wins out. Faced with this harsh reality, the best solution is thought by many to be to publish fewer, longer, higher-quality papers. *I believe, in fact, that it is more efficient to publish less papers and better papers, and target the most relevant journals, even their IFs are not the highest. It makes you more visible to your community (thanks to the quality of the paper) and your paper is published more quickly if you spent enough time to consolidate and improve it. It is the strategy I applied, and I believe that it worked because I obtained a position.*

There is very little interest in/or enthusiasm for publishing in open access mega-journals and concerns about quality, including that of the flagbearer, PLOS One, is largely the reason for this. ECRs like the idea of social media outreach, which is publishing informal research presentations to a wider audience, such as through blogs, but the reality is that few practiced this because of a lack of time and absence of reputational reward.

Open access publishing

Despite deeply held ECR commitments – that have not been shaken over the years – to the idea of open access and being increasingly marshalled towards open access by mandates, entrenched reputational concerns and appraisal systems frustrate increased practice.

This is a scholarly activity where, arguably, there is the greatest debate. Attitudes and practices are generally not in harmony, with the latter lagging behind and often dependent on whether the ECR is speaking as an author or reader. They have a split personality. However, ECRs have become much more aware

of open access during the life of the project: *I am more aware of open practices in my new university. It is more common among ECRs, but it comes from the seniors. I see taking back control for academics as important in dissemination.*

ECRs are keen in principle, and the direction of scholarly travel fits well with their millennial beliefs concerning transparency, access, and sharing, but they recognize that, in practice, for purposes of grants, promotion, and tenure, open access ranks low in the criteria when seeking a journal in which to publish a research paper. It is a particular problem in subject fields where there are no “serious” or high impact factor open access journals. This is especially the case for engineering and physics. There has been an increasing commitment to open access on the part of ECRs, but this does not necessarily lead to more open access publishing. Indeed, in some countries, such as Poland, there appears to be a greater emphasis on publishing in top journals recognized by high impact factors and usually not open access, but sometimes hybrid.

Open access is universally considered to be ethical by ECRs, but the positive, practical arguments voiced in favour of open access is that it offers greater visibility and access and hence encourages more sharing. There is also a sense that, because of this, they will obtain more citations, which is, importantly, the very currency by which they are appraised and obtain a reputation.

The big issue, other than impact factor, is the costs that have to be paid to publish, and given that ECRs are at the bottom of the pecking order when it comes to handing out article processing charges (APCs), it is understandable that this issue rises to the surface so frequently in interviews. Even when funding is available, they feel it could be better spent elsewhere. Attitudes to repositories and green open access (and it is not readily understood) is at best lukewarm, and as a consequence, there appears to be little practice undertaken, and mandates, where known, are thought to be an irritant.

Open access, then, is a difficult scholarly aspect for ECRs to navigate as this quote shows only too well: *OA is fine as long as the journal is ISI rated, reputable and the university will pay the article processing fees but you need to make sure that it is not listed in Beall's list. Not predatory. But the university will not pay if you want to make the article free in an OA (hybrid) journal.*

Open data

Open data is in transition, and inevitably, practices differ between disciplines and countries, and individual researchers sometimes change their views and courses of action.

As with open access, ECRs are generally welcoming of data sharing, which is thought good for reproducibility and verification purposes and, hence, good for science. However, a large number do not share, at least not until after publication of the all-important paper that will cement their reputation. There is a clash between the wish to use their evidence first (in papers in high ranking journals) and the obligation to openness. One ECR put the dilemma like this: ‘The data is my most valuable asset, I paid money and worked hard for it, why open it freely to others?’

Another reason for low take-up of open data initiatives is the lack of incentives: 'The current reward system does not provide much incentive for Open Science (open data sharing and green open access), and to change the current culture is hard'.

Publishing in data journals has not taken on, and there was a general agreement that if you wished to make your research more visible, accessible, and more reproducible, having a good and detailed Methods section in your article was the best way.

Sharing

Social media have enabled sharing to such an extent that there can be a conflict between ECRs and publishers. Widespread sharing is delivering research dividends, but concerns remain about hard-earned results being given away in a competitive environment, where being original and the first to publish counts.

Sharing as an attitude was picked out by ECRs as a mark of their millennial birth right. In principle (and with reservations – see below), sharing research information is thought to be a very good thing for the great majority of ECRs; they really warm to it and would like to share more than they do, but time and publisher restrictions get in the way. In principle, they are certainly less worried about sharing their research because they might lose their competitive edge. It is also one of those rare scholarly areas where ECRs distinguished their attitude and practices from their mentors who were said to be more cautious when it came to sharing.

Face-to-face meetings and conferences still have a special place here, but so do social scholarly networks, such as ResearchGate, which is essentially all about sharing – too freely for some publishers. Much of that sharing concerns uploading papers to networks: *I am aware of the need to disseminate my research work. I realise that if I want to cooperate with other scientists, I have to share my work and information about me in the social media.* Sharing outside the community of researchers, to policy makers for instance, is especially aspirational among European ECRs.

The sharing of published papers poses few problems for ECRs; they are concerned about sharing unpublished papers, interim results, data, and ideas because of competition and the possibility of people stealing information/ideas and making them their own. They want protection and recognition until the research is published. When it comes to the sharing of published papers, the main issue is publishers' copyright restrictions for uploading full texts to the social media. ECRs are not generally aware of publisher initiatives to reduce 'friction' where article sharing takes place among academics and to help make it a consistent and user-friendly experience for researchers.

Collaboration

ECRs are very aware of the career advancement and knowledge transfer value of collaborations, and as a consequence, it is exhibiting the fastest growth in interest of all the scholarly activities. It is probably the one scholarly aspect of which they cannot obtain enough.

Sharing and collaboration go together, so it comes as no surprise that collaboration is much sought after too. It is undertaken for a whole range of ECR-friendly reasons, including: reputational building (i.e. more citations received); co-authorship opportunities proffered; and general improvement in research quality. There is also the fact that it is easier to undertake these days thanks to the scholarly social networks, as explained by one ECR: 'We are lucky, our generation has many tools and places to identify people and teams with whom collaborations are possible'.

For all these reasons, collaboration is increasing with every researcher, especially Chinese ECRs, working very hard towards collaborating with internationally famous research groups, teams, and projects. They are encouraged to do this by governments and funders.

As with sharing, there is always the worry that other researchers will steal their work, and there is also the recognition that collaboration can lead to more work, but the rewards are thought to far outweigh the risks.

Social media

Social media do deliver for ECRs and are being used for a wide range of mainstream scholarly activities, such as sharing, networking, outreach, and showcasing.

There is less unanimity here in terms of what ECRs think and do. This is partly because of cultural and language differences, most obviously in the case of China where Chinese-specific services, such as WeChat, are dominant ('I use WeChat for scholarly communication almost every day') and partly because there are differences in opinion over the value of prioritizing (but not using) the social media.

There is, however, a general feeling among many ECRs that social media are useful for showcasing their research, providing them with greater visibility and thus obtaining a bigger audience for it and, hopefully attracting more citations as a result. Some also mentioned the fact that it enables them to collaborate and feel closer to fellow researchers. One Chinese researcher took things to another level by saying he or she felt that the social media 'inspired' him or her and made him or her more 'creative'. There is also an increasing number of researchers who are stressing the importance of social media exposing them to different voices, viewpoints, and ways of presenting their research and also as a place for taking the pulse of their field of study. 'Through social media, I can spot research trends and communicate instantly. By using the social media, I can get different sounds'. ResearchGate is often mentioned in these contexts and is thus hence very popular with ECRs. Of the generic services used, Twitter, Facebook, LinkedIn, and YouTube have now been joined by a new generation of platforms – WhatsApp and Telegram Web, an instant messaging system, and Instagram, mainly in Malaysia and China.

ECRs, however, generally do not have sufficient time to utilize social media or are provided with sufficient reputational reward that will help them to prioritize it. Yes, voices of concern are still heard, but there are far fewer of them than at the

beginning of the project. There is, however, one thing they steadfastly will not do and that is cite the social media. That is going too far.

Summing up, it does seem that social media have come of age. Reputational concerns and time pressures associated with paper publishing do stand in the way of greater use. However, there is no real evidence to believe that, in time, social media will take over the role of the more traditional means of communication, such as paper publishing, and so transform it.

Metrics

Increasing measurement, especially by citations, is a key characteristic of the scholarly communications system and is used widely in appraisals and for reputational purposes. ECRs are uncomfortable with the dominance of metrics but are not highly critical of them.

Metrics are seemingly a rock-solid aspect of scholarly communications, and by metrics, ECRs really mean citation-based ones, which provide the principal means of evaluating the formal system, its papers, and users. We have here another case of ECRs not necessarily liking a scholarly aspect, more a case of having to put up with it. Maybe that is why, in conversations with ECRs, they appeared bored when asked about them, seldom coming to the defence of citations by pointing to their attractions. Although, plainly, they are well aware of their importance to their appraisals, careers, and reputation. As one ECR said: 'Citations are everything and everything is related to citation'. Ironically, even in the face of potential competition from altmetrics, citations are becoming even more entrenched, and this is because universities view citations as a measure of research excellence and the means to improve their world rankings.

There is, however, a growing interest in altmetrics among ECRs, but despite what purveyors of altmetrics may say, they do not regard them so much as measures of research impact but more social impact – a gauge of popularity, reach, interest, and curiosity. *Impact is all about citations and/or utility for society, altmetrics on the other hand just tell us about curiosity.* What ECRs, however, are waking up to is the fact that altmetrics could lead to more citations: 'Citations are the important research measure, but obtaining good altmetrics might mean that you will get citations in the future'.

Recent interest is, no doubt, fuelled by the fact that a few universities (in China especially) are providing ECRs credit for their altmetric scores and that national policies in Spain and France, for instance, are being formulated to take into account social impact scores of ECRs. Finally, there is some discomfort (no more than that) that ECRs are part of an increasingly metric-driven world in which metrics are not actually measuring quality or impact and narrowing academic choices.

Impact

After collaboration, research impact is the activity that shows the greatest growth in ECR interest, but impact is still fundamentally all about citations.

For ECRs, every scholarly activity has a goal, which is to increase their competitive edge in order to obtain that prized secure position. This is why there is so much interest in making an impact, and it is, in fact, the fastest growing scholarly aspect in terms of increased positive interest and practice for ECRs. This is, no doubt, fuelled by national policies, such as the UK's Research Evaluation Framework and blandishments from publishers and social scholarly networks platforms claiming they can offer evidence of 'impact'. ECRs view impact as largely influencing one's peers, and as mentioned, success in this is largely determined by citations. Less so, it is also seen to be successfully reaching out to policymakers and the general public, 'I am more conscious of the importance of the dissemination of our research to a general public in order to wake up the interest of the generations about science. So, with the new technologies is very easy to make this so'. Spanish ECRs are particularly passionate about making an impact by reaching out: 'To build bridges with society is a must'. Most would concur with this but do not feel qualified to engage in outreach activities, and because of the demands of paper publishing, they simply did not have the time to engage. ECRs are awaiting government initiatives to enable them to undertake these activities and obtain the requisite reward or reputational advantage.

Ethical practices

There is recognition that ethics is a widespread problem, with the reason being that the reputational system is so demanding with respect to paper publishing and predatory journals, which open the door to wider unethical behaviour.

Because there is no consensus here as to what is considered unethical and because of its sensitivity, it is a difficult scholarly aspect to question ECRs about, but there is a recognition that there is a problem. It is widely felt by ECRs that more misconduct has occurred in recent years, and this has been combated by the introduction of more and better ethical procedures/training and calls to integrate ethics standard in all aspects of the research and publishing cycle. ECRs are more knowledgeable because of the likes of Committee on Publication Ethics (COPE), Retraction Watch, and the Scholarly Kitchen blog, but there is a concern that the existing policing systems are being overwhelmed by the sheer volume of unethical behaviour.

The blame for rising levels of unethical behaviour is levelled variously at predatory publishers, open (unpoliced) science, journal editors attempting to raise the impact factor of their journals, reviewers with vested interests, and the pressures placed on researchers to publish.

There is, however, a marked difference in opinion between the English-language countries and the others, with ECRs from UK and USA exhibiting very little interest or concern about ethical behaviour. This could be explained by two factors: (1) coming from mature scholarly communication countries, ECRs believe – possibly complacently so – that their house is in order, and (b) they believe their own publishing behaviour is ethical and are not particularly interested in generalizations about what others are doing.

ECRs come up with different examples of what they constitute as being unethical, and the following list illustrates the wide range of activities identified:

- 'Copying others' work' (plagiarism)
- 'Paying someone to ghost write our paper'
- 'Publishing things that are not true' (fake data)
- 'Not accurately portraying data' (covering up negative results)
- 'Stealing people's ideas or gazumping someone based on a presentation' (at a conference)
- 'Using data more than once in different papers'
- 'Anything that could deceive others'
- 'To give too many citations to friends, colleagues and not provide citations to competitors'.

Reputation

The global reputational system remains largely unyielding despite the clear concerns of ECRs. This, in turn, frustrates changes taking place elsewhere in the system.

Reputation and journals go together, with one reinforcing the other, and this creates a very powerful and monopolistic partnership, which ought to suit publishers. The widely acknowledged reality is that publishing papers is the single most important factor in establishing and maintaining reputation. It is even more specific than that with one ECR speaking for many when stating that reputation was actually delivered by *Publishing in Q1 and Q2 journals in JCR for STM. Scopus is OK for the social sciences*. The real debate seems to be whether it was WoS or Scopus that provided the better accreditation. So embedded is this belief that one ECR who moved to an industrial research position where they are not allowed to publish worried how she was going to get recognition now.

Although a minority of ECRs said they would rather be judged by the quality of their research rather than by the journals in which they published, it was thought that many more felt this but could not admit it. They were just faced with the reality that: 'there isn't another robust system for research quality control and evaluation'.

A few ECRs mentioned that data creation, peer review, and some social media activities should contribute towards reputation. However, they are very much voices in the wilderness, and the global reputational system appears unyielding, with no big breakthroughs occurring. With competition never fiercer for work and funds, as long as the international yardstick for research success is measured by publishing papers, nothing seems likely to change.

Libraries

Libraries have an invisible presence and, with respect to the activity that counts most with ECRs – Access to the full-text paper – they face real challenges from a borderless and more competitive information environment.

There has been little change in the attitudes and behaviours of ECRs regarding libraries over the period of study and far less than is the case for *all* the other scholarly aspects documented here. For many ECRs, libraries are an invisible presence, largely used (often unconsciously) for one purpose and one purpose only – obtaining remote access to the full text of papers. This role is appreciated, however, because it is so central to ECRs' research life. Of course, open access and platforms, such as Sci-Hub and ResearchGate, are chipping away at their territory. As one ECR said 'I think Google Scholar will replace the research library in the future'. There is thought to be some value of the library as a space, what a French ECR called a "*disconnected place*" where they can write papers undistracted. A Malaysian ECR dwelt a little more on what this would mean: 'My university library provides lovely environment for study, comfortable sofa, coffee machine, free computers and printing service. The role of library has already been changed to environment provider rather than information service provider'. Of course, for this to happen, libraries would have to give up on their research function, which is doubtful that they would want to do.

There is more recognition of the worth of libraries in the USA/UK, perhaps because of their attempts to attract researchers by other means: advising ECRs about open access and copyright, offering institutional repositories for their publications, and helping them with their ORCID entries and institutional research submissions. Overall, however, libraries are clearly languishing in many countries and are in danger of being disintermediated by the likes of scholarly social networks and, indeed, by open access publishing.

Publishers

Publishers are viewed as intermediaries and organizers of the validated scholarly communication system and, of course, hold the keys to the reputational system – those high impact factor journals.

While libraries appear to be threatened by ECR attitudes and behaviours, the same cannot be said about publishers, possibly as a surprise to some. They (the traditional ones, anyway) are talked about as being 'strategic, indispensable and trustworthy'. This is because of their central role in the journal publishing process. One ECR pointed out that if publisher functions were removed, researchers would have much more work on their hands, and they did not want that to happen. That is not to say there are no complaints or issues. There are still the perennial criticisms that ECRs and other researchers do all the work, and yet, publishers insist on charging 'such high APCs'. But these criticisms are only strongly held by French ECRs. Oddly enough, throughout this project, there have been relatively few complaints about the publishing processes, its slowness, and such like.

CONCLUSION (AND THE FUTURE)

Listening and learning from the current experiences of ECRs, it would seem that some scholarly mechanisms and facilities

generally work well for them: discovery and access, authorship practices, peer review, sharing collaboration, and publishers are in this category. However, the closely connected aspects of reputation, publishing strategies, metrics, and impact do not work so well. These scholarly aspects also have a knock-on effect and cause some tensions and dissatisfaction with other scholarly aspects – social media, open access publishing, and open data. Of the rest, libraries' research role appears to be under threat, and as for ethical behaviour, there is a divide between those who believe it is a growing problem and those who do not.

At the end of the questioning on individual scholarly aspects, ECRs were asked to reflect on what they had said and what they thought the future would hold and whether they foresaw any scholarly transformations occurring in the next 5 years. It turned out that ECRs are uncertain about this as they were of some of the individual scholarly aspects. There was no consensus as to whether there will be change, what will be changed, and whether ECRs will have a key role to play in that change. However, few envisaged that transformational change would take place in the 5-year period (aside from French ECRs), although most thought that change should happen, and it probably will occur in a slow and gradual manner. There is little sign, however, of them relinquishing their beliefs and ambitions with regard to sharing, openness, and transparency, and just maybe, their call to measure reputation and have a more comprehensive and sensible impact will be heard in an increasing metric-driven world.

LIMITATIONS

The sample of ECRs is relatively small and not necessarily representative of the population as a whole. In order to summarize the huge amount of collected data, we had to generalize.

STATEMENT OF INFORMED CONSENT

Interviewees provided informed consent to participate in the study and were given the opportunity to review their data. Guarantees of privacy, security, and anonymization were provided.

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